

Computational Statistics for Data Science (CSDS)

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Instructor

Soumya Ray
soumya.ray@iss.nthu.edu.tw
Office: 850 TSMC Building

Teaching Assistants

(contact on MS Teams only)
Edgar Duron
Vy (范青翠薇)
Angel (任翊瑄)

Prerequisite knowledge:

You must have completed at least one statistics class before taking this course. This class focuses on the computational approaches to statistics, and only lightly reviews fundamental statistics topics.

No prior programming knowledge is required but students with basic coding experience will have an advantage. Beginners must be willing to learn basic programming through homework tutorials. We will teach intermediate programming techniques in class that build on the homework tutorials.

Course Description:

This class covers computational methods for statistics. You will learn to develop tools that support decision-making, and produce reproducible analyses and reports.

Computational focus: We will use *computational methods of programming and algorithms* to overcome limitations of traditional statistical methods. We will learn to reshape data, run simulations, discover dimensions in data, utilize machine learning, and create complex models.

Methodological focus: We will use statistics to *understand, explain, and predict scientific and business phenomena* to inform decision making and conduct research. We will describe and visualize data, create statistical models from domain knowledge, test our ideas against data; predict new outcomes; and guard against fallacious use of statistics.

Skill focus: We will learn to *write data analytics code in R on par with industry standards*. Students will implement their own algorithms, write code that is highly readable and reusable, produce highly performant code, create custom visualizations, and apply different styles of analytic programming.

Software Tools:

Students will use the RStudio or Visual Studio Code to write code in the R programming language. *Our tools are all free and open-source.* Using R requires learning basic concepts in programming and maintaining code, which we will teach in class. If you wish to familiarize yourself with these tools before class begins, please start with the tutorials at <http://swirlstats.com>

Ethics Statement on Generative Artificial Intelligence and other Programming Resources:

*In accordance with the published Guidelines for Collaboration, Co-learning, and Cultivation of Artificial Intelligence Competencies in University Education, this course adopts the following policy: **Conditionally open.***

Grounded in the principles of transparency and responsibility, this course encourages students to leverage Generative AI (GenAI) like ChatGPT and Copilot in some of the assignments, which will be noted on the assignment. This means that students may use GenAI when permitted, but:

1. Must briefly explain how AI was used (of if not used) for every homework assignment.
2. Code copied from GenAI must be properly explained if it deviates from what is taught in class.
3. Must not submit any GenAI code that they do not fully understand.

We expect students to carefully study, verify, and suitably alter their GAI assisted solutions to match homework requirements and reflect their genuine learning. If code is discovered that was not properly understood and adapted, or if the use of GenAI or other resources was not disclosed, instructors have the right to reevaluate the assignment or withhold scores.

Students enrolled in this course agree to the above ethics statement.

Grading Policy

Assignments: You will receive grades every week that count towards your total final grade

- *Individual homework assignments (50%):* weekly homework reports will be *anonymously peer-reviewed* by two other students chosen at random; TAs will assign scores based on peer-reviewers' suggestions. You may offer a *rebuttal* if you feel that you deserved a better grade, and our teaching team will arbitrate.
- *Peer review (24%):* you thoughtfully review two students' assignments each week after class.
- *Tutorials and quizzes (13%):* occasional interactive tutorials and quizzes on readings.
- *Engagement (13%):* your attendance will be recorded and graded from Week 3 onwards. Meaningful online discussion can be credited towards class absences.

Extra Credit: You can earn extra points throughout the semester:

- *Outstanding Homework:* your peers may grant you an extra point for noteworthy reports
- *Assistance Credit:* if your peers mention your help on submitted assignments.
- *Online discussion:* if you ask or answer questions in-class, or in online discussions on MS Teams.

Reference Material:

Lecture slides will be provided before every class and made available online at our class website.

Students will be given extra material from videos, research papers, practitioners' blogs, and more.

Course Topics

Computational Perspective

Computation and Statistics

Learning Computation
Exploration, Inference, and Prediction
Our Tools: R and Rstudio

Industry: Peer Review and Social Learning

Tutorial: Swirl to Learn R

Description and Simulation

Kernel Density Plots / Histograms
Simulating Distributions
Inferential Statistics

Coding: Conceptualizing Variables

Coding: Simulating Data from Distributions

Computing: Binary Representation of Numbers

Tutorial: Swirl to Learn R

Computational Intervals

Functions and Iterations
Describing Distributions
Confidence Intervals
Resampling

Coding: Writing Your Own Functions

Coding: Functional and Vectorized Iteration

Coding: Performance Benchmarking

Simulation: Sampling Statistics

Tutorial: Swirl to Learn R

Computational Tests

Bootstrapping

Review of Descriptives
Classical Hypothesis Testing
Bootstrapping the Alternative

Reading: Random Walks

Reading: Android Malware Detector

Statistics: Rescaling Data

Nonparametric Testing

Bootstrapped Hypothesis Testing
Empirical Distributions and Power

*Simulation: Null and Alternative
Statistics: Type I, Type II Errors
Data: Website Performance*

Permutation Tests

Reshaping Data
Permutation of Data Samples
Wilcoxon Test: Permutation vs. Sum of Ranks

*Tutorial: Swirl to Learn R
Data: Verizon Customer Service*

Multigroup Tests

Normality and Quantiles – the QQ Plot
ANOVA: Parametric Test for Multiple Groups
Kruskal Wallis: Nonparametric Test of Independent Groups

*Coding: How to Choose R Packages
Statistics: Familywise Errors
Data: Media Experiment*

Inferring Relationships in Data

Data Similarity

Data as Vectors
Similarity: Cosine, Correlation
Item-Item Collaborative Filtering

*Reading: Collaborative filtering at
Amazon
Statistics: p-hacking and Frequentist
Mistakes
Statistics: Dot Products
Reading: Collaborative Filtering at
Amazon*

Dual Perspectives of Linear Regression

Review of Linear Regression
Geometric Representation of Regression
Linear Algebraic Representation of Regression

*Simulation: Interactive Regression
Data: Cars Dataset
Videos: Essence of Linear Algebra
Reading: Amazon Retrospective on
Recommender Systems*

Applied Regression

The Hat Matrix
Diagnosing and Managing Non-Linearity
Diagnosing and Managing Multi-Collinearity

*Statistics: Stepwise-VIF
Videos: Essence of Linear Algebra*

Moderation and Mediation

The Contingency Perspective as Moderation
Partial Orthogonalization
Bootstrapped Test of Indirect Effects

Videos: Essence of Linear Algebra

Data Dimensions and Latent Variables

Composites and Components

Principal Components
Transforming Dimensions
Reducing Dimensions

*Data: Online Security Survey
Data: Decathlon Athletics
Simulation: Interactive PCA*

Principal Components Analysis

Composite Variables
Composites vs. Factors
Component Rotation as Perspective

*Coding: Anonymous Functions
Coding: Pipes & Forward Moving Code
Statistics: Parallel Analysis*

Structural Equation Modeling

Structural Models

Composite Structural Models

Common Factor Structural Models

Coding: SEMinR package by class alumni

Coding: Domain-Specific Languages

Coding: Functional Currying

Coding: Open-Source Communities

Machine Learning

Predictions

Out-of-sample Predictions

Split-sample Testing

k-Fold Cross Validation

Statistics: Polynomial Regression

Statistics: leave-one-out cross validation

Machine Learning: Decision Trees

Ensemble Predictions

Stable vs. Unstable Algorithms

Bagging Algorithms

Boosting Algorithms

Data: Insurance Dataset

Coding: Updating Estimated Models

Coding: expand.grid vs. nested for-loops

Validation and Conclusions

Hyperparameter Tuning

Validation Sets

What's Next?

Coding: High Performing Data.Table package

Coding: RStudio Server

Coding: Shiny Web Applications

Coding: Matrix package for sparse matrices